Project Design Phase -I
Problem Solution Fit

Team ID	LTVIP2026TMIDS36743
Project Name	visualization tool for electric vehicle charge and range analysis

Problem Solution Fit :

In our project, **"Visualization Tool for Electric Vehicle Charge and Range Analysis"**, we observed that EV users and fleet operators often face difficulty in accurately monitoring battery charge levels, predicting driving range, and analyzing energy consumption patterns. Inconsistent range estimation, inefficient charging management, and lack of real-time insights can lead to range anxiety, poor trip planning, and reduced vehicle performance.

These challenges are commonly caused by improper data monitoring, lack of predictive analysis, inefficient battery usage tracking, and absence of structured visualization systems. As a result, EV performance analysis becomes complex, less reliable, and time-consuming, which can impact decision-making for both individual users and organizations.

To overcome these challenges, our project focuses on developing an interactive visualization system for effective EV charge and range analysis, including:

Battery Performance Monitoring:

Tracking real-time battery charge percentage, voltage, current, and temperature.

Analyzing charge–discharge cycles to evaluate battery health and efficiency.

Range Prediction & Analysis:

Developing predictive models to estimate remaining driving range based on speed, terrain, and driving behavior.

Visualizing energy consumption per kilometer for better trip planning.

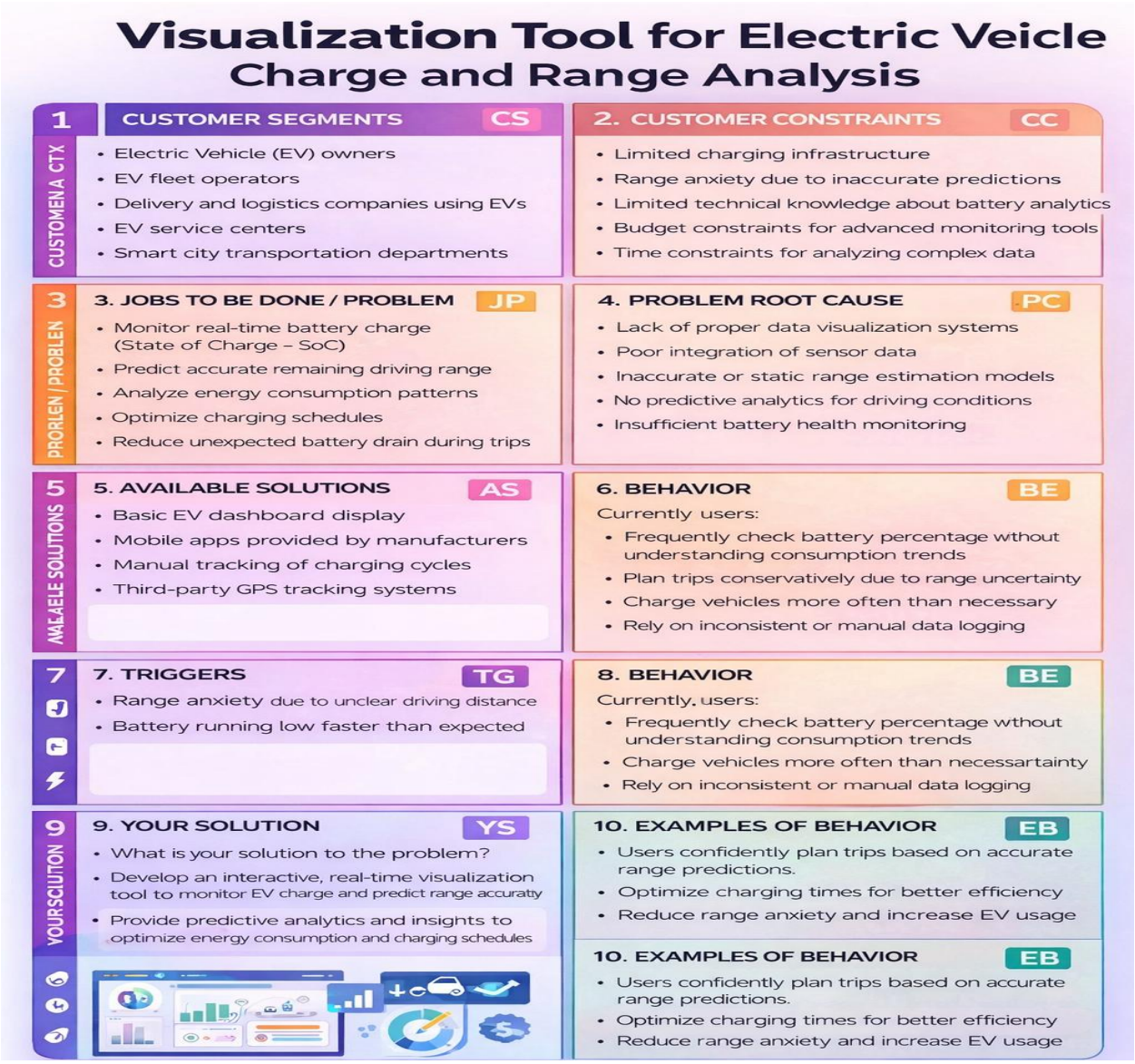
Data Optimization & Processing:

Filtering and cleaning sensor data to ensure accurate range estimation.

Pre-processing large EV datasets to improve dashboard performance and responsiveness.

Dashboard Design Optimization:

Designing interactive dashboards to display:

- State of Charge (SoC)
- Estimated Remaining Range
- Energy Consumption Trends
- Charging Time Analysis

Using efficient chart types and minimal visual clutter to ensure fast rendering and user-friendly analysis.